EXPERIMENT NO:04

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Normalize data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create ANN using backpropagation
```

```
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation='relu',
    solver='adam',
    max_iter=2000,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Predict output
y_pred = model.predict(X_test)

# Display accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", round(accuracy * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Graph 1: Training loss
plt.figure(figsize=(7, 4))
plt.plot(model.loss_curve_, marker='.')
plt.title("ANN Backpropagation Training Loss")
plt.xlabel("Iterations")
plt.ylabel("Loss")
```

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plt.grid(True)
plt.show()

# Graph 2: Confusion matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))
plt.imshow(cm, cmap='Blues')
plt.title("Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.xticks([0, 1, 2], iris.target_names)
plt.yticks([0, 1, 2], iris.target_names)
plt.colorbar()

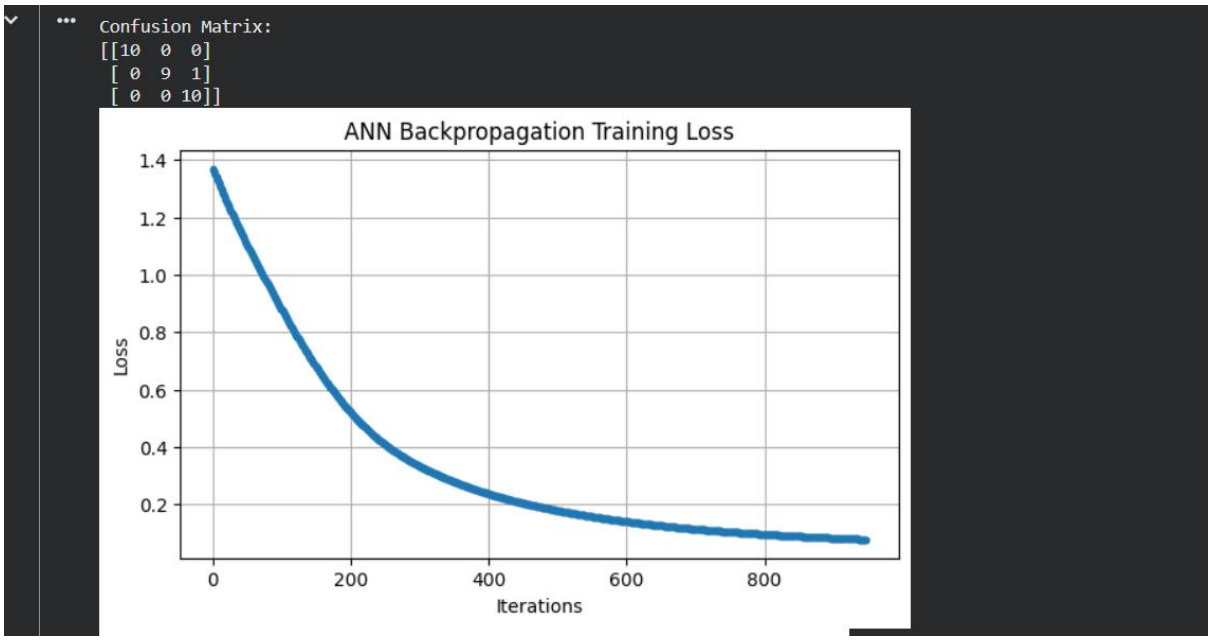
for i in range(3):
    for j in range(3):
        plt.text(j, i, cm[i, j], ha='center', va='center')

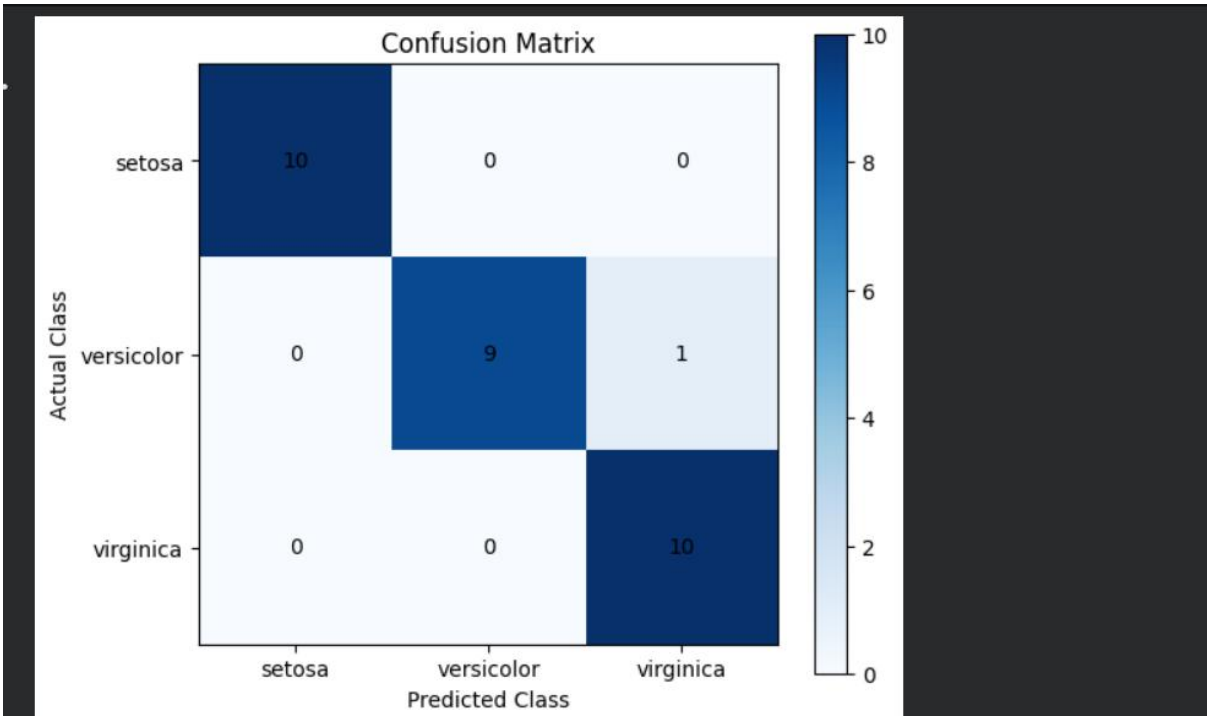
plt.tight_layout()
plt.show()

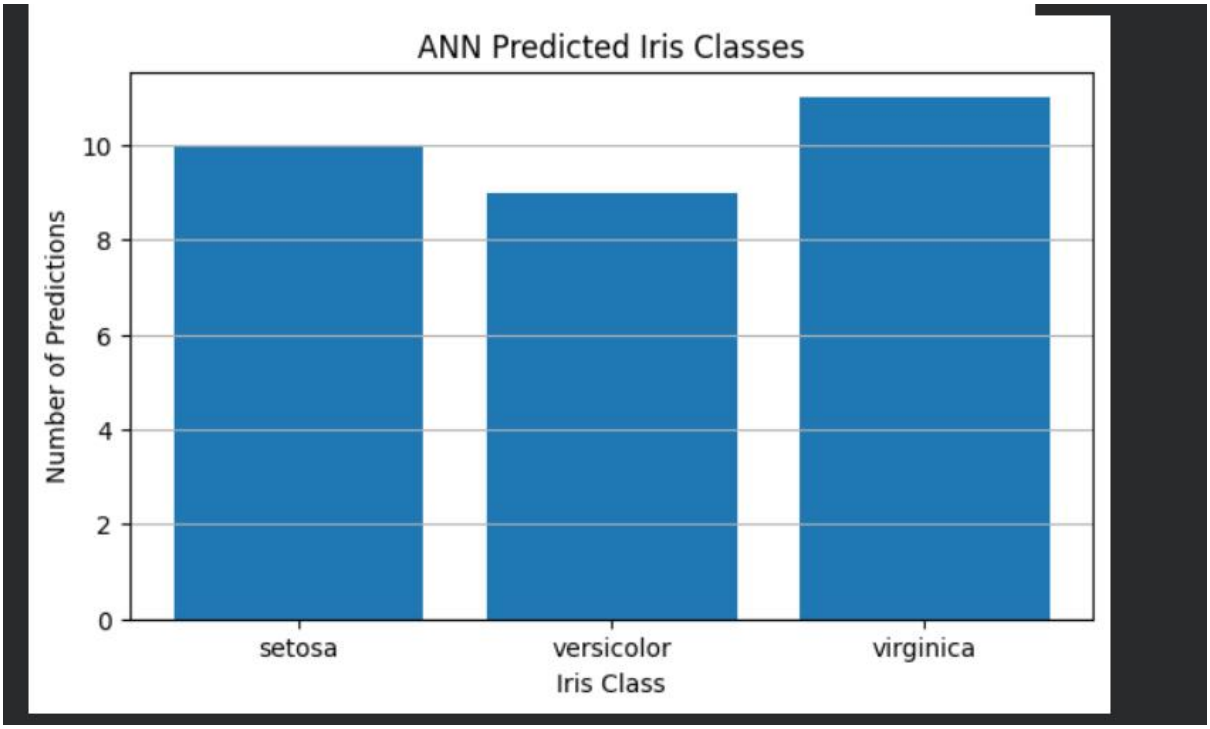
# Graph 3: Predicted class distribution
plt.figure(figsize=(7, 4))
plt.bar(
    iris.target_names,
```

```
    np.bincount(y_pred, minlength=3)
)
plt.title("ANN Predicted Iris Classes")
plt.xlabel("Iris Class")
plt.ylabel("Number of Predictions")
plt.grid(axis='y')
plt.show()
```

OUTPUT:




Confusion Matrix
setosa
versicolor
virginica
Actual Class
10
0
0
0
9
1
0
0
10
setosa
versicolor
virginica
Predicted Class
10
8
6
4
2
0


ANN Predicted Iris Classes
Number of Predictions
10
8
6
4
2
0
setosa
versicolor
virginica
Iris Class